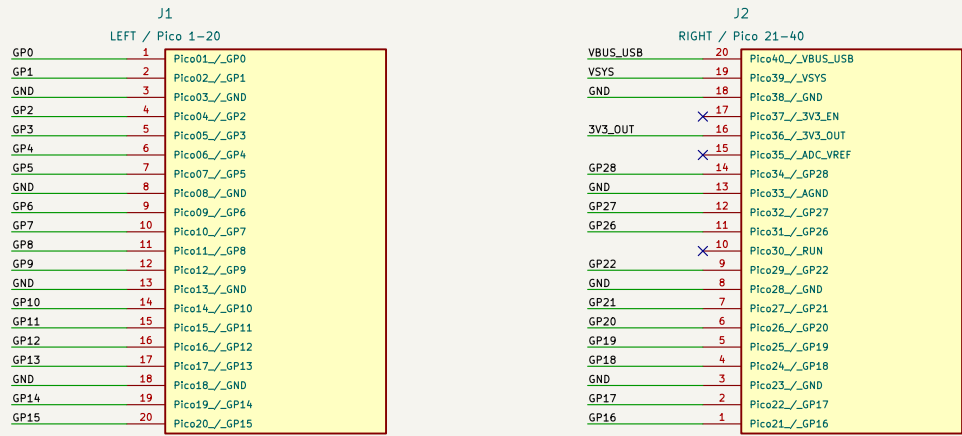


## MAKE MUSIC – PROVISIONAL PICO CARRIER

Editable KiCad 10 schematic | Module order, parts, power and mechanical fit require primary integration

### REMOVABLE PICO: TOP VIEW, USB AT TOP

Selected Pico H SC0917: original RP2040, fitted male headers. Socket pitch2.54mm / rows17.78mm; fit trial pending.



J1 pin 1 = Pico 1 (USB end), J2 pin 1 = Pico 21 (opposite USB).

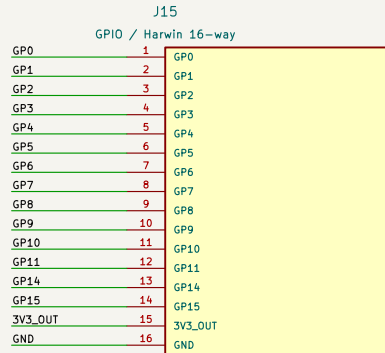
J2 pad 20 = Pico 40 at USB end: rotate J2 footprint 180 degrees relative to J1 in placement.

NC: Pico 30 RUN, 35 ADC\_VREF, 37 3V3\_EN, On-module defaults retained.

Pico 33 AGND tied to GND for digital touch use. GP23–25/29 are internal, not expansion.

SSQ candidate: 1.02 mm drills / 1.52 mm pads. Verify finished holes and male–header fit.

### FREE GPIO EXPANSION – CARRIER ORDER DEFINED



GP0/GP1: optional I2C SDA/SCL.

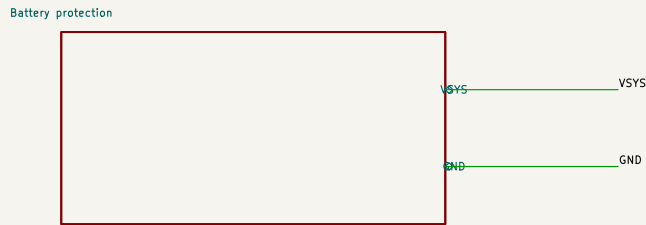
No I2C pull–ups fitted until bus/device choice.

### REVIEW ONLY – NOT FOR FABRICATION

ERC checks carrier wiring only. Cell PCM, module electronics and assembled fault/load/fit behavior remain unverified. See power sheet.

Sources: Raspberry Pi original Pico datasheet, Figure 2 pin allocation; see ../reviews/schematic.md.

### FUSED BATTERY POWER – SHEET 2



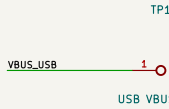
File: battery\_power.kicad\_sch

H1 JST harness: circuit1 protected positive, circuit2 GND.

F1 + U2 replace the old carrier series D1.

Switch OFF isolates battery; USB may still power Pico.

Stop playing before source changes: reboot may occur.

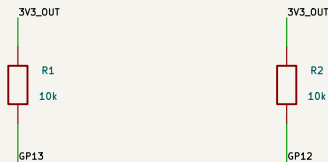


External charging only: cell/holder fit still pending.

ERC checks carrier wiring only. Cell PCM, module electronics and assembled fault/load/fit behavior remain unverified. See power sheet.

Sources: Raspberry Pi original Pico datasheet, Figure 2 pin allocation; see ../reviews/schematic.md.

### BUZZER RESET BIAS



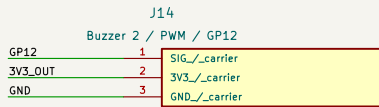
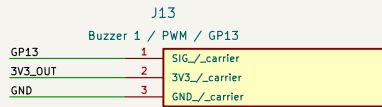
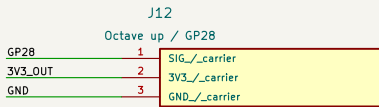
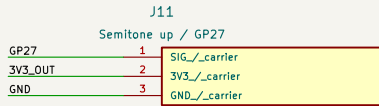
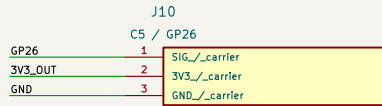
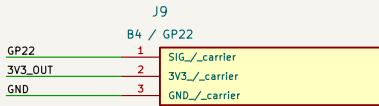
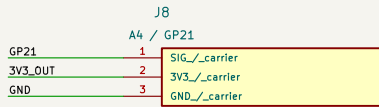
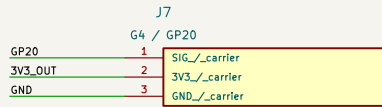
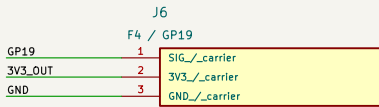
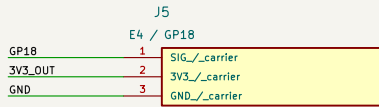
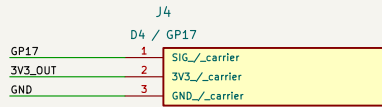
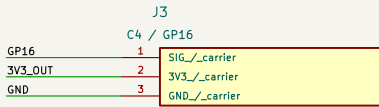
10k 1% each; nominal 0.66mA total when both LOW.

Carrier pull–ups require connected 3V3/GND/cables.

### TEN TOUCH / TWO BUZZER CARRIER CABLE HEADERS

J3–J14 carrier–defined order: 1=SIG, 2=3V3, 3=GND. Map cable to module by signal label.

Harwin carrier header footprints assigned. No direct module–mount/hole geometry; see cable contract.



### SIGNAL / POWER LIMITS TO RESOLVE

Touch: TTP223 at 3.3 V, direct mode (TOG=0). Firmware may adapt to confirmed active–high/low.

ST0238 low–level driver: R1/R2 bias GP13/12 HIGH at reset; firmware HIGH idle still required.

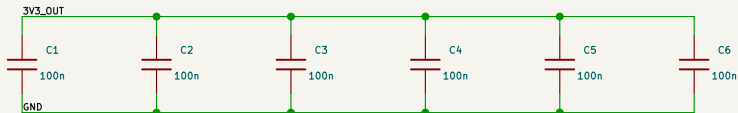
GP12 + GP13 share a PWM slice/frequency. Baseline firmware plays both at the same pitch.

Budget Pico 3V3\_OUT current including 10 touch modules, 2 buzzer drivers and expansion loads.

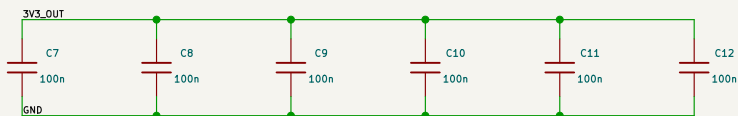
J1/J2: SSQ–120–01–G–5 candidate assigned; mating fit, USB/BOOTSEL and battery clearances TBD.

Reference may be active–low (AHLB=1). Confirm actual polarity/pin order before firmware changes.

C1–C12: 100nF / 50V X7R per carrier header



C1/J3 .. C12/J14: place locally; short GND return.



Prototype review – fit and bench validation pending

Sheet: /  
File: make\_music.kicad\_sch

**Title: Make Music – provisional Pico carrier**

Size: A3 Date: 2026–10–08

KiCad E.D.A. 10.0.6

Rev: P4

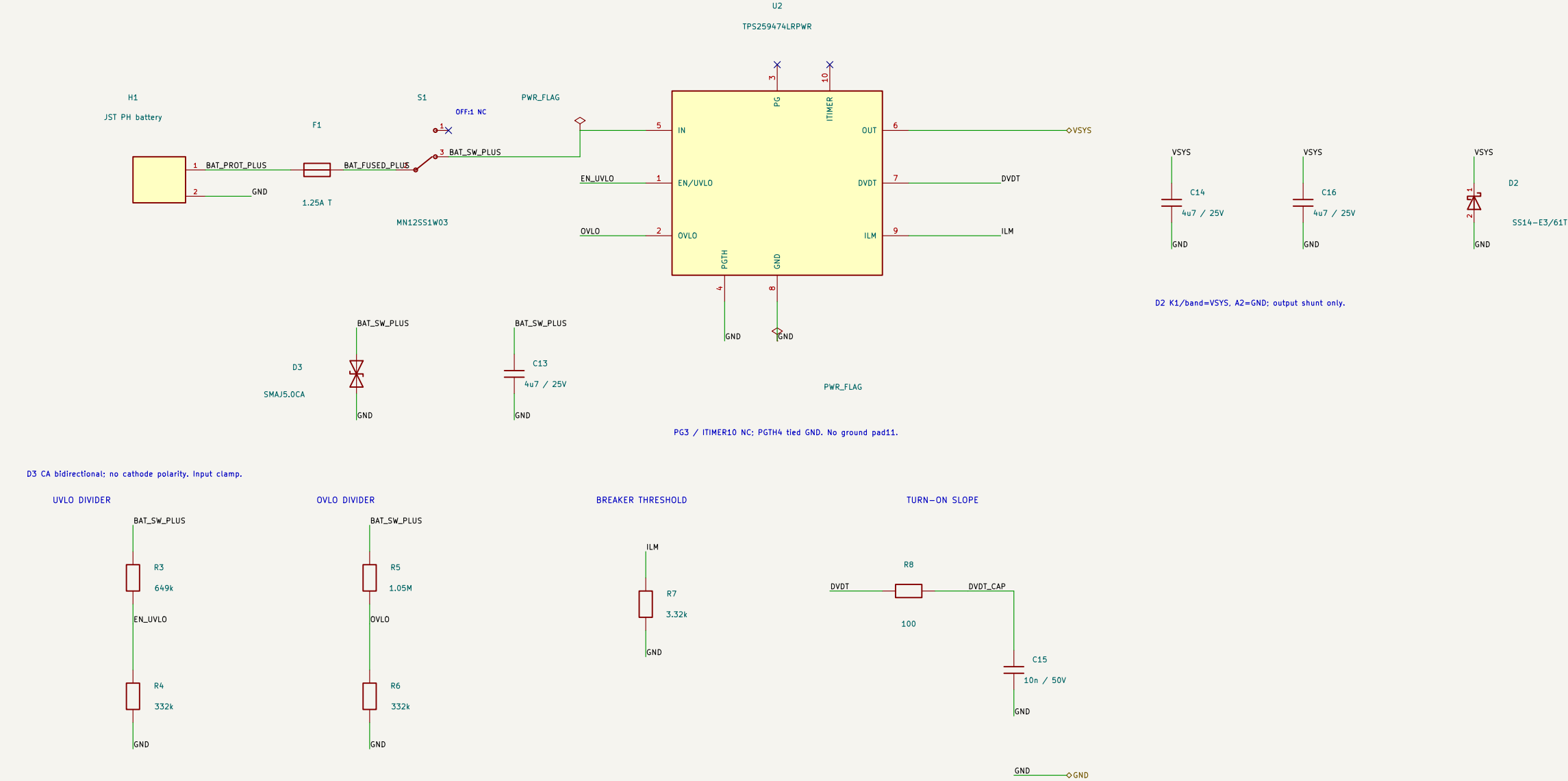
Id: 1/2

BATTERY BRANCH – FUSED, REVERSE–PROTECTED, EXTERNAL CHARGE ONLY

Prototype circuit from power PR #18. Physical fit, current, fault transients and source changes require bench validation.

ENTRY / BACKUP FUSE / SWITCH / LATCH–OFF eFUSE

LOCAL NONPOLAR CERAMICS / SHUNT CLAMPS



DESIGN LIMITS: branch <=0.70A; total3V3 <=350mA including Pico / external <=250mA; ambient0–40C.

All VSYS capacitance including Pico <=100uF; effective output ceramics >1uF. No polymer hold–up parts.

UVLO nominal3.221V off /3.546V restart at IN; breaker nominal1.004A. Tolerance/1S fault response need bench checks.

Stop playing before USB/battery changes; reboot may occur. S1 OFF / USB removal before service; external cell charging only.

F1 first at entry, before branches; 5mm–wide /35um copper approach lands per power contract. Keep unfused harness short/insulated.

Local C13 at IN/GND. C14/C16 at OUT/GND; short clamp/control loops. RPW IN5/OUT6 are large power contacts.

PWR\_FLAGS declare external switched supply and return for ERC only; they do not add hardware or prove live power.

Reviewed candidate footprints and circuit are not fabrication approval. See power–integration.md and power/branch–protection.md.

Prototype review – fit and bench validation pending

Sheet: /Battery protection/  
File: battery\_power.kicad\_sch

**Title: Make Music – fused battery power**

Size: A3 Date: 2026–10–08  
KiCad E.D.A. 10.0.6

Rev: P4  
Id: 2/2